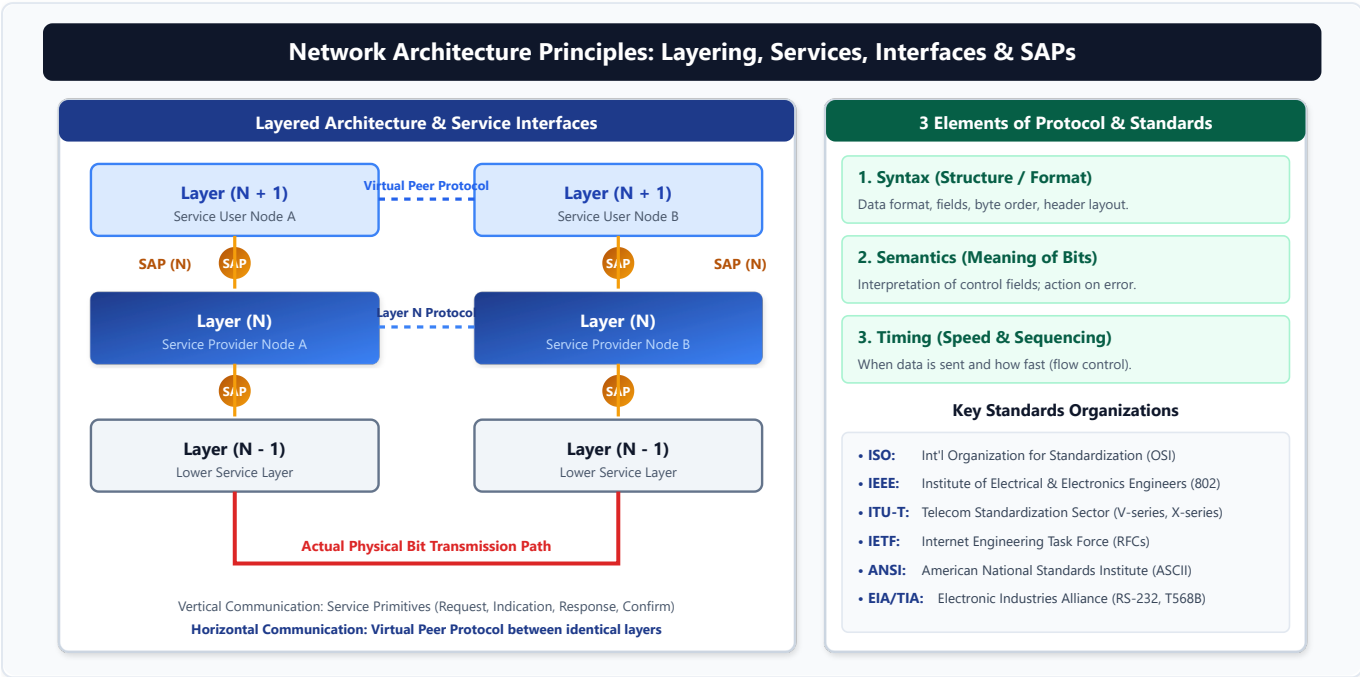


Module 04: Protocols, Standards and Layering Principles

AKTU BCS603 Focus: Syntax, Semantics, Timing, De Facto vs De Jure, Standards bodies (ISO, IEEE, IETF, ITU-T), Layering concepts, SAPs, and Service Primitives.



1. 3 Key Pillars of Every Protocol

- **Syntax:** How data is structured into bit patterns (Header format, field offsets, flags).
- **Semantics:** Interpretation and actionable meaning assigned to bit patterns (e.g., error alert vs acknowledgment).
- **Timing:** Transmission synchronization, rate-matching flow control, and timeout thresholds.

2. Service vs Protocol Distinction

Dimension	Service (Vertical)	Protocol (Horizontal)
Interaction Axis	Between Layer (N) and Layer (N-1) on the <i>same</i> machine	Between Peer Layer (N) on Machine A and Peer Layer (N) on Machine B
Interface Point	Service Access Point (SAP) using Primitives (Request, Indication, Response, Confirm)	Network transmission medium carrying Protocol Data Units (PDUs)
Visibility	Internal to operating system stack / hardware drivers	Visible over wire as network packet headers

3. Standards Organizations Cheat Sheet

- **IEEE 802 Committee:** 802.3 (Ethernet), 802.11 (Wi-Fi), 802.15 (Bluetooth).
- **IETF (RFCs):** RFC 791 (IPv4), RFC 793 (TCP), RFC 2616 (HTTP/1.1).
- **ISO:** ISO 7498 Open Systems Interconnection (OSI Reference Model).

AKTU Critical Difference: De Facto means standards set by widespread custom without law (e.g. TCP/IP, USB Type-C popularity); De Jure means formal standards legalized by sanctioned treaty bodies (e.g. ISO 7498, IEEE specs).